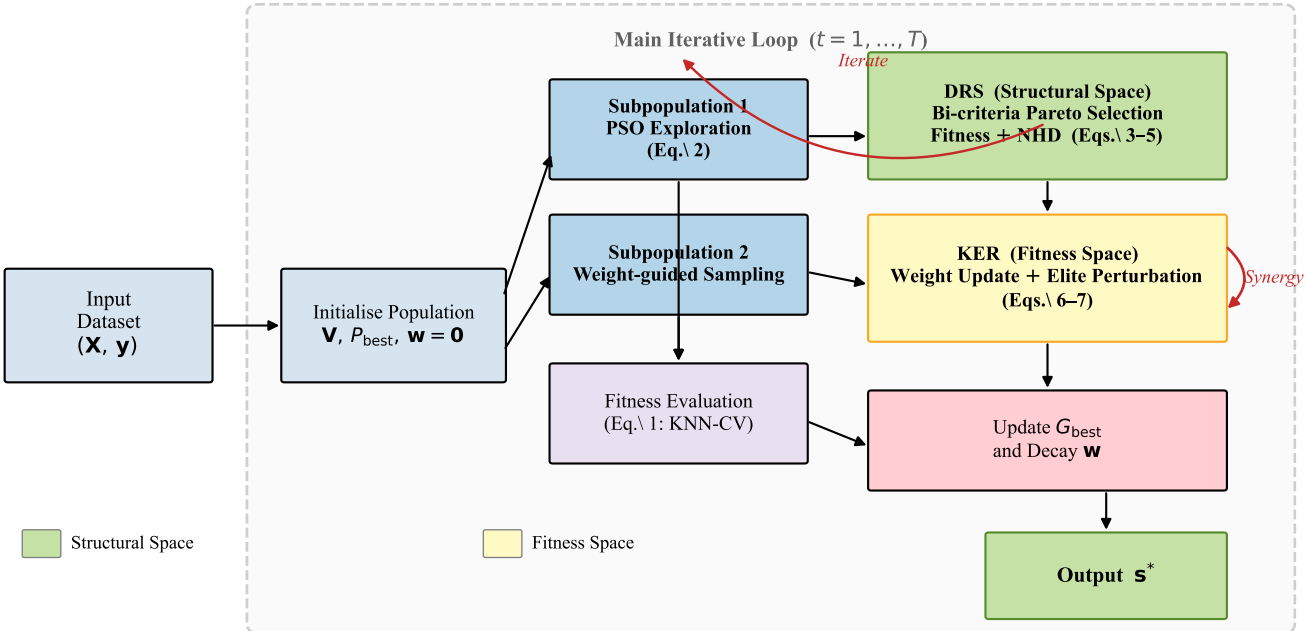

Main Iterative Loop ($t = 1, \ldots, T$)
Iterate
Input Dataset ($\mathbf{X}$, $\mathbf{y}$)
Initialise Population $\mathbf{V}$, P_{best}, $\mathbf{w} = \mathbf{0}$
Subpopulation 1 PSO Exploration (Eq.\ 2)
Subpopulation 2 Weight-guided Sampling
Fitness Evaluation (Eq.\ 1: KNN-CV)
DRS (Structural Space) Bi-criteria Pareto Selection Fitness + NHD (Eqs.\ 3–5)
KER (Fitness Space) Weight Update + Elite Perturbation (Eqs.\ 6–7)
Synergy
Update G_{best} and Decay $\mathbf{w}$
Output $\mathbf{s}^*$
Structural Space
Fitness Space